

CMP-15 - Verification Service

Evaluate explicit claims against declared criteria, methods, environments, evidence, uncertainty, and reproducibility requirements.

Authority and state ownership

Book I: Decisions 24-29, 35, 40-43, 47-50, and 58. Book II: Chapters 17, 35. Authoritative state: Verification Plans, Verification Results, claim-evaluation state, fidelity and confidence records. Explicitly non-owned: Evidence Objects, Certification decisions, simulation runtime, release approval.

Critical invariant

Verification is scoped and evidence-based; it never converts an unsupported claim into authority or certification.

Logical interfaces

CMP-15-IF-01 / Command / Start Verification

Command `Start Verification` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-15-IF-02 / Command / Record Verification Result

Command `Record Verification Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-15-IF-03 / Command / Invalidate Result

Command `Invalidate Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-15-IF-04 / Command / Request Reproduction

Command `Request Reproduction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-15-IF-05 / Query / Get Verification Result

Query `Get Verification Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-15-IF-06 / Query / Explain Confidence

Query `Explain Confidence` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-15-IF-07 / Query / List Defeaters

Query `List Defeaters` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-15-IF-08 / Event / Verification Started

Event `Verification Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-15-IF-09 / Event / Claim Verified

Event `Claim Verified` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-15-IF-10 / Event / Claim Falsified

Event `Claim Falsified` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-15-IF-11 / Event / Result Invalidated

Event `Result Invalidated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Proposed > Running > Verified, Falsified, Inconclusive, or Invalidated. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: criterion drift, environment mismatch, evidence substitution, irreproducible result, confidence inflation. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-15-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-15-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-15-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-15-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-15-TST-005 - Critical invariant. Verification is scoped and evidence-based; it never converts an unsupported claim into authority or certification. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-15-TST-006 - Failure containment. Inject criterion drift, environment mismatch, evidence substitution, irreproducible result, confidence inflation. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-15-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-15-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-15-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-15-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.